

IC 4011, whose truth table is shown in Table IV.

Auto-search method. In this

process, the number of pins of the IC to be checked is entered first. The device then starts manifesting all

possible input signals to the IC and takes back its response for each possible input. If a response matches the output of a particular IC in its database, then it declares that IC as good (Figs 7 and 8).

Manual-checking method.

In this method, the IC number is entered first (for example, 4011 as shown in Fig. 9). On continuation, basic detail of that IC is displayed (Fig. 10). At the start of the checking process, an option for truth table is provided for the user (Fig. 11). For viewing the truth tables, this option must be selected. At the next stage, the MCU initialises the signal-processing task.

In case of this specific NAND gate IC 4011, the MCU provides 5V supply to pin 14 and 0V to pin 7. As this IC has four NAND gates, each of these is checked one by one. The MCU provides the necessary combination of inputs to each gate as per the truth table (Table IV) and takes back outputs from the IC (4011) as its input.

Then, by comparing these observed results with expected results as per IC specifications, the MCU yields its conclusion on that particu-

TABLE III
DIFFERENT FUNCTIONS

Functions	Basic operations
void loop()	Executes again and again
void auto_manual()	For selecting the mode of testing (automatic/manual)
int auto_search()	For automatic searching of the IC and giving its working status
int welcome()	For displaying a simple welcome screen
int help()	For displaying some important instructions
int test()	For accepting IC number and calling the exact function for the specific IC
long int accept(char ch)	For accepting different inputs (for example, IC number, time intervals and binary numbers etc)
int ic_quad_2(int num)	For checking those ICs that have four 2-input gates (for example, 4011)
int ic_triple_3(int num)	For checking those ICs that have three 3-input gates (for example, 4073)
int ic_74_quad_2(long int num)	For checking some 74 series ICs that have four 2-input gates (for example, 7400)
int ic_hex_inv(int num)	For checking an IC with six inverting gates
int ic_4029()	For checking counter IC 4029
int ic_dual_4(int num)	For checking those ICs that have two 4-input gates (for example, 4082)
int ic_oct_1(int num)	For checking an IC with an 8-input gate
int ic_74393()	For checking counter IC 74393
int ic_hex_16pin(int num)	For checking some hex-buffers and hex inverters (for example, 4050 and 4049)
int ic_4000()	For checking a dual 3-input NOR gate (for example, 4000)
int ic_d_flip(int num)	For checking a dual D flip-flop (for example, 7474, 7479)
int ic_jk_flip(int num)	For checking a dual JK flip-flop (for example, 7476 and 7473)
int dec2bin(int n)	For converting numbers from decimal to binary

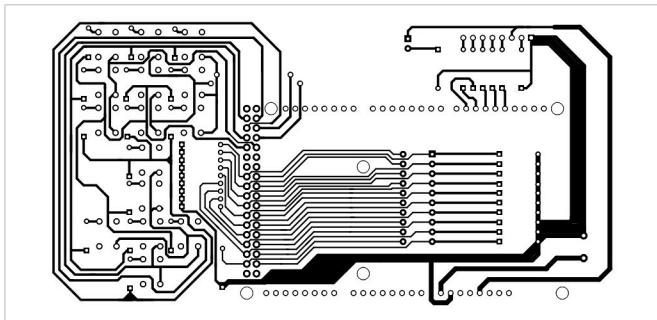


Fig. 4: Actual-size PCB pattern of the digital IC tester